

LECTURE 6

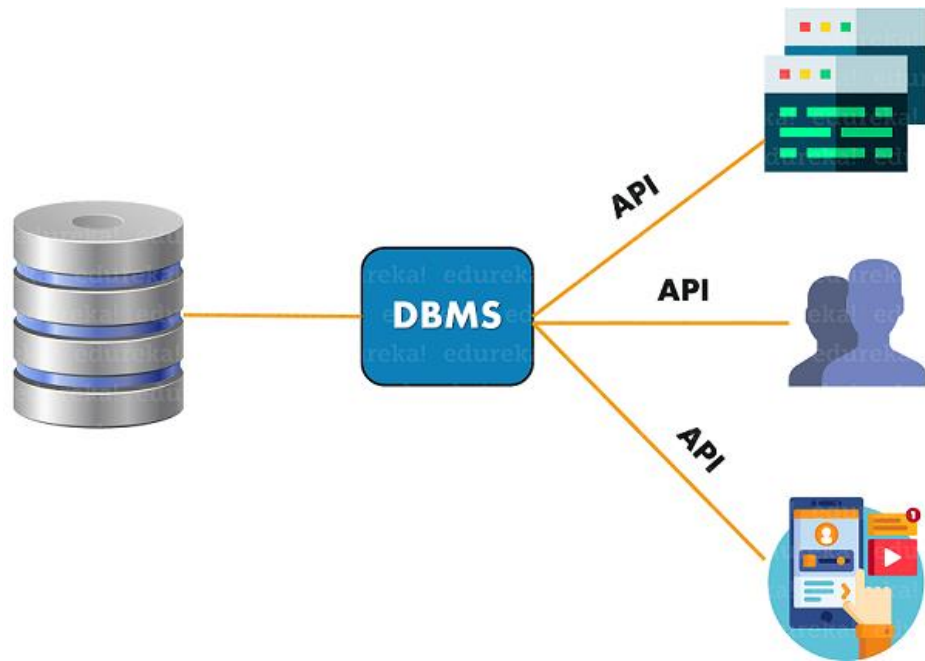
Introduction to Database and SQL Statements

WEB DEVELOPMENT 2

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What Is A Database?

Databases are used to **store** and **organize** large amounts of data in a **structured** way. They are designed to **manage** and **handle** large volumes of **information** efficiently and effectively, making it easy to **retrieve**, **update**, and **delete** data as needed.





DATABASE ENGINES

MySQL database server



Software methods to connect database

- PHPMYADMIN
- MYSQL workbench
- Shell / CMD



SQL statements

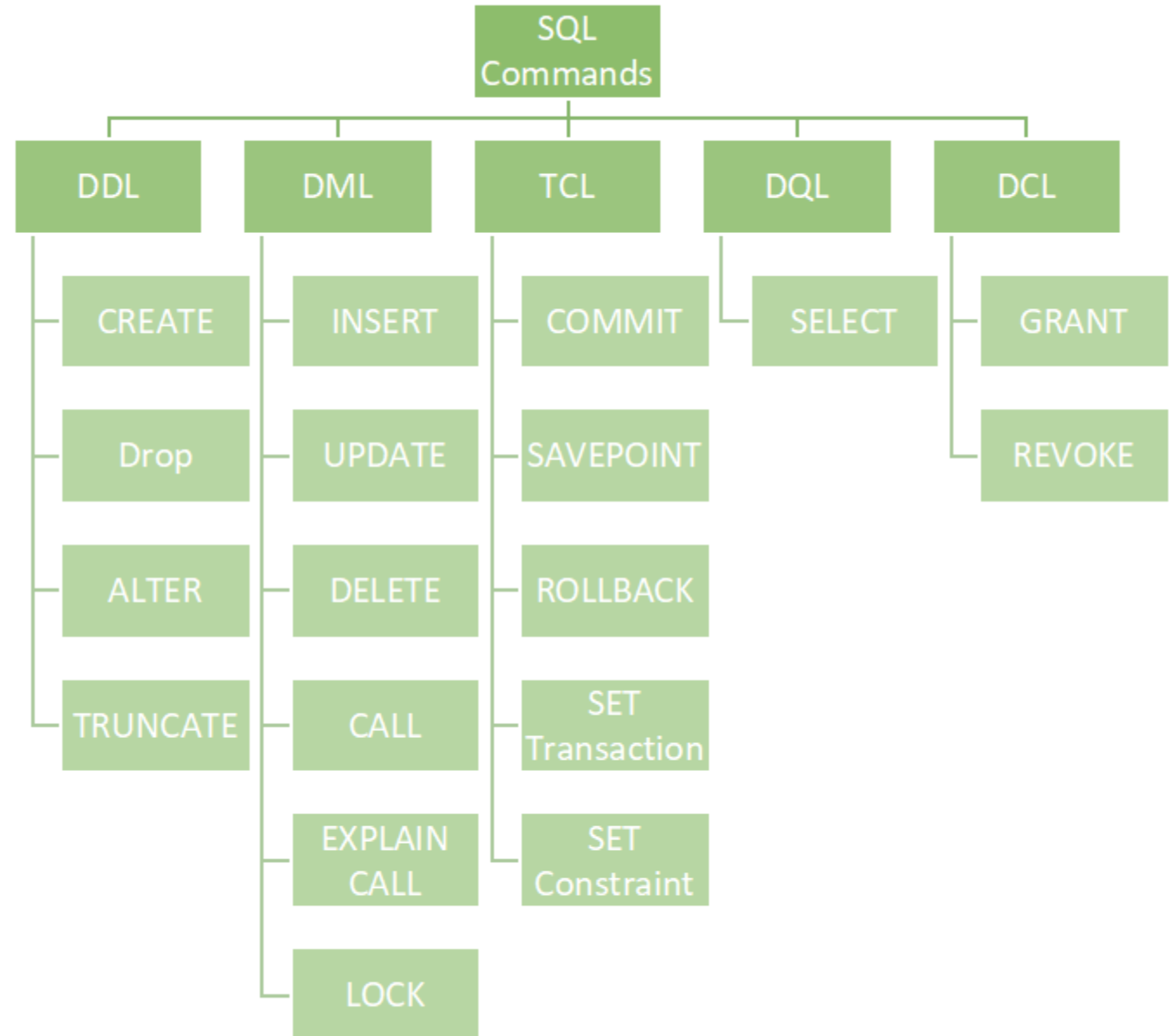
DDL – Data Definition Language

DQL – Data Query Language

DML – Data Manipulation Language

DCL – Data Control Language

TCL – Transaction Control Language



Creating a new Database in SQL

```
CREATE DATABASE Webdevelopment;
```

Verifying the Database

```
SHOW DATABASES;
```

SQL Drop Database

```
DROP DATABASE database_name;
```

SQL Select database

USE DATABASE_NAME;

Verifying the Database

SHOW DATABASES;

SQL CREATE TABLE Statement

```
CREATE TABLE Customer(  
    CustomerID INT PRIMARY KEY,  
    CustomerName VARCHAR(50),  
    LastName VARCHAR(50),  
    Country VARCHAR(50),  
    Age int(2),  
    Phone int(10)  
);
```

DROP TABLE

DROP TABLE;

Adding a Column to a Table:

ALTER TABLE tablename ADD columnname datatype;

Renaming a Table:

rename table name1 to name2;

Truncating a Table:

TRUNCATE TABLE tablename;

Retrieve Specific Columns from Users Table

```
SELECT Username, Email FROM Users;
```

Retrieve all data

```
Select * from table_name
```

Insert Data into Users Table

-- Inserting specific data into the Users table

```
INSERT INTO Users (Username, Email, Password) VALUES  
( 'user3', 'user3@example.com', 'password3'),  
( 'user4', 'user4@example.com', 'password4');
```

Exercise: Creating a Basic Blog Database

Create Users Table

Create a table named "Users" with the following columns:

UserID (integer)

Username (varchar(50))

Email (varchar(100))

Password (varchar(255))

Exercise: Creating a Basic Blog Database

Create Posts Table

Create a table named "Posts" with the following columns:

- PostID (integer)
- Title (varchar(255))
- Content (text)
- CreatedAt (timestamp)
- UserID (integer)

Exercise: Creating a Basic Blog Database

Insert Sample Data Insert the following data into the "Users" table:

- UserID: 1, Username: 'user1', Email: 'user1@example.com', Password: 'password1'
- UserID: 2, Username: 'user2', Email: 'user2@example.com', Password: 'password2'

Insert the following data into the "Posts" table:

- PostID: 1, Title: 'First Post', Content: 'This is the content of the first post.', CreatedAt: [current timestamp], UserID: 1
- PostID: 2, Title: 'Second Post', Content: 'This is the content of the second post.', CreatedAt: [current timestamp], UserID: 2

Exercise: Creating a Basic Blog Database

Retrieve Data Write SQL queries to retrieve the following information:

- List all users from the "Users" table.
- List all posts from the "Posts" table.

Answer

Create Users Table

```
CREATE TABLE Users (  
    UserID INT,  
    Username  
    VARCHAR(50),  
    Email VARCHAR(100),  
    Password  
    VARCHAR(255)  
);
```

Create Posts Table

```
CREATE TABLE Posts (  
    PostID INT,  
    Title VARCHAR(255),  
    Content TEXT,  
    CreatedAt TIMESTAMP,  
    UserID INT  
);
```

Answer: Insert Sample Data

```
INSERT INTO Users (UserID, Username, Email, Password) VALUES  
(1, 'user1', 'user1@example.com', 'password1'),  
(2, 'user2', 'user2@example.com', 'password2');
```

```
INSERT INTO Posts (PostID, Title, Content, CreatedAt, UserID) VALUES  
(1, 'First Post', 'This is the content of the first post.', CURRENT_TIMESTAMP, 1),  
(2, 'Second Post', 'This is the content of the second post.', CURRENT_TIMESTAMP, 2);
```

Answer: Retrieve Data

-- List all users from the Users table

```
SELECT * FROM Users;
```

-- List all posts from the Posts table

```
SELECT * FROM Posts;
```

- Retrieve only the Username and Email columns from the Users table

```
SELECT Username, Email FROM Users;
```

-- Retrieve only the Title and CreatedAt columns from the Posts table

```
SELECT Title, CreatedAt FROM Posts;
```


resources

- <https://www.w3schools.com/>